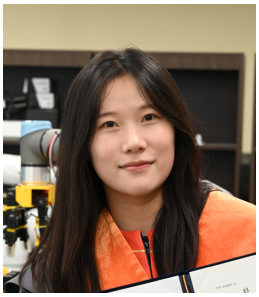
# YoungChae Son



I am **YoungChae Son**, a Ph.D. student at Dongguk University in Seoul, working with Prof. Soo-Chul Lim in the Interactive Robotics Lab. My research focuses on robot learning and planning with vision-language foundation models, including Vision-Language Models (VLMs) and Vision-Language-Action (VLA) models. I am particularly interested in enabling robots to make adaptive decisions in real-world environments where task-relevant information may be incomplete, ambiguous, or change during execution. My work explores multimodal perception, task planning, and adaptive manipulation, with an emphasis on integrating and prioritizing sensory observations for long-horizon execution on real robotic systems.

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## Education

2025 – Present
**Dongguk University** — *Seoul, South Korea*
*Ph.D. in Mechanical Engineering*
Interactive Robotics Lab | **Advisor:** Prof. Soo-Chul Lim

2023 – 2025
**Dongguk University** — *Seoul, South Korea*
*M.S. in Mechanical Engineering*
Interactive Robotics Lab | **Advisor:** Prof. Soo-Chul Lim

2021 – 2023
**Tech University of Korea** — *Siheung, South Korea*
*B.S. in Mechatronics Engineering*
(2018 – 2021) Seoul National University of Education, Self-Education in Home Economics

## Publications

**Journal Article** 2026
**See Selectively, Act Adaptively: Dual-Level Structural Decomposition for Bimanual Robot Manipulation**
Yoon-Ji Choi, **Young-Chae Son**, Soo-Chul Lim*
*Under Review*

**Journal Article** 2026
**Task-Aware Modality Selection for Multimodal VLA in Robotic Manipulation**
**Young-Chae Son**, Jung-Woo Lee, Yoon-Ji Choi, Dae-Kwan Ko, Soo-Chul Lim*
*Under Review*

**Journal Article** 2026
**ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making**
**Young-Chae Son**, Dae-Kwan Ko, Yoon-Ji Choi, Soo-Chul Lim*
*IEEE Robotics and Automation Letters (RA-L)*

**Conference Paper** 2026
**A Nonmagnetic Stepper Motor for the Automatic Evaluation of Magnetic Field Uniformity in Helmholtz Coils**
**Young-Chae Son**, Sung-Jung Joo, Po-Gyu Park, Changjin Yun, Soo-Chul Lim*, Hyung-Kew Lee*
*IEEE Conference on Precision Electromagnetic Measurements (CPEM)*

**Journal Article** 2026
**MulPlanLM: Multimodal Robotic Task Planning with Vision-Language Models and Physical Feedback**
**Young-Chae Son**, Dong-Han Lee, Soo-Chul Lim*
*Intelligent Service Robotics*

**Conference Paper** 2022
**The Remote Robotic Arm Control Using Inertial Sensor and Image Recognition**
Ki-Jun Han, **Young-Chae Son**, Yang-Hee Lee*
*Korean Society for Precision Engineering (KSPE)*

**Conference Poster** 2022
**High SNR and High Frame Rate Analog Front-End ICs with Active Stylus for Capacitive Touch Screen Panels**
**Young-Chae Son**, Jae-Sung Ahn*
*Europe-Korea Conference on Science and Technology (EKC), Marseille, France*

## Selected Research Projects

Jul. 2025 – Present
**Development of Intelligent Autonomous Manipulation Technology for Humanoids with Reduced Dependency on Real-World Data**
*Humanoid Manipulation, Data-Efficient Learning, Robot Foundation Models* | **Agency:** Ministry of Science and ICT (MSIT)

Mar. 2025 – Present
**Robot Motion Generation AI Based on Multimodal Vision/Tactile Information Driven by Language Model**
*Multimodal Robot Learning, Vision/Tactile Perception, Language Models* | **Agency:** National Research Foundation of Korea (NRF)

Aug. 2024 – Dec. 2024
**Collecting Large-Scale Robot Manipulation Data in Physical Environment**
*Robot Data Collection, Manipulation, NVIDIA Isaac Gym* | **Agency:** National Information Society Agency (NIA)

## Patents

Apr. 2026
**Multimodal Robot System for Task-Context-Based Selective Sensory Utilization**
Korean Patent Application No. 10-2026-0073734

Apr. 2026
**Method, Apparatus, System, and Computer-Readable Recording Medium for Robot Control Using Thermal Image Information**
Korean Patent Application No. 10-2026-0067117

## Presentations

Oct. 2026
**ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, PA, USA

Jun. 2026
**Selective Perception for Robot: Task-Aware Attention in Multimodal VLA**
Robot Learning Research Group Workshop, Korea Institute of Industrial Technology (KITECH)

Feb. 2026
**ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making**
Locomotion & Manipulation / Robot Learning Research Groups, Korea Robotics Conference (KRoC), Pyeongchang

## Honors & Awards

Sep. 2025 – Present
**Global Outstanding Talent Scholarship**
Full tuition waiver for Ph.D., Dongguk University

2026
**Best Paper Award**
BK21 FOUR AIMS Center Winter Conference, Dongguk University

2022
**Grand Prize (1st Place), Transfer Student Mentoring Testimonial**
Tech University of Korea

2022
**Grand Prize (1st Place), English Presentation Contest**
Tech University of Korea

## Teaching & Work Experience

Spring '24, '25, '26
**Teaching Assistant, Autonomous Robotics**
Dongguk University | *Autonomous driving lab using ROS 2 and TurtleBot3 Burger*

Fall 2024
**Teaching Assistant, Computational Mechanics**
Dongguk University

Fall 2023
**Teaching Assistant, Fluid Mechanics**
Dongguk University

Oct. 2022 – Jun. 2023
**Interactive Robotics Lab, Dongguk University**
Undergraduate Research Intern